

# Momentum — Modern AI Habit Tracker

## Product & Engineering Plan

### Vision

Momentum adalah habit tracker modern dengan fokus pada clean UI, dopamine-driven UX, dan AI-powered behavioral insights. Project ini dirancang bukan hanya sebagai CRUD app, tetapi sebagai portfolio yang terasa seperti produk startup nyata.

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### Core Differentiation

Dibanding habit tracker biasa, Momentum memiliki:

- AI consistency analysis
  - Offline-first architecture
  - Realtime sync multi-device
  - Advanced heatmap visualization
  - Product psychology onboarding
  - Smooth optimistic UI interactions
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### Modern Tech Stack

#### Frontend

- Nuxt 4
- Vue 3 + TypeScript
- TailwindCSS v4
- shadcn-vue
- VueUse

#### Backend

- Nuxt Server API (H3)
- Drizzle ORM
- PostgreSQL

#### Authentication

- Better Auth

#### Observability

- PostHog
  - Sentry
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### Business Perspective

Project ini diposisikan sebagai micro SaaS modern. Target user:

- Students

- Developers
- Productivity enthusiasts
- Self-improvement audience

Monetization ideas:

- Premium themes
- AI insights subscription
- Advanced analytics
- Team habit tracking

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## Improved Product Features

### 1. Smart Habit System

User dapat membuat habit dengan recurring schedule, category, dan target behavior.

### 2. AI Consistency Insights

AI menganalisa pola habit gagal dan memberi rekomendasi.

### 3. Offline First Experience

User tetap bisa menggunakan app tanpa internet dan sync otomatis saat online.

### 4. Realtime Multi Device Sync

Progress langsung update antar device.

### 5. Advanced Heatmap

GitHub-style contribution graph dengan animation dan yearly rewind.

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## Architecture

Client (Nuxt App)

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API Layer (H3 Server Routes)

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Service Layer

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Drizzle ORM

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PostgreSQL

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## Folder Structure

```
app/  
components/  
pages/  
layouts/  
composables/  
server/  
■■■ api/  
■■■ db/  
■■■ services/  
■■■ utils/
```

types/

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## Engineering Improvements

Untuk meningkatkan nilai portfolio di mata recruiter dan senior engineer:

- Optimistic UI updates
  - Offline sync queue
  - IndexedDB local persistence
  - Rate limiting
  - Database indexing
  - Unit testing (Vitest)
  - E2E testing (Playwright)
  - Monitoring & analytics integration
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## User Flow

Landing Page → Register → Onboarding → Create First Habit → Daily Check-in → Analytics Dashboard → AI Insights

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## MVP Timeline

### Week 1

Project setup, auth, database schema

### Week 2

Habit CRUD + dashboard

### Week 3

Heatmap, charts, animations

### Week 4

AI insights, optimization, deployment

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## Why This Portfolio Is Strong

Project ini menunjukkan:

- Fullstack engineering
  - Product thinking
  - Modern architecture
  - AI integration
  - UX polish
  - Scalability awareness
  - Startup-oriented mindset
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# Context7 MCP Workflow

Recommended MCP integrations:

- Context7 MCP → latest framework documentation
- Filesystem MCP → AI project awareness
- PostgreSQL MCP → schema understanding
- GitHub MCP → repo & PR automation

Recommended AI coding tools:

- Cursor
  - Claude Code
  - Windsurf
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